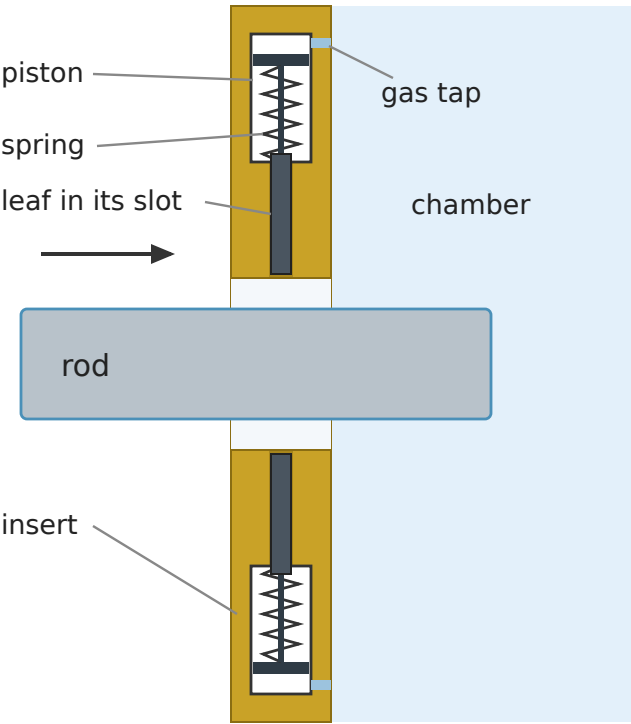
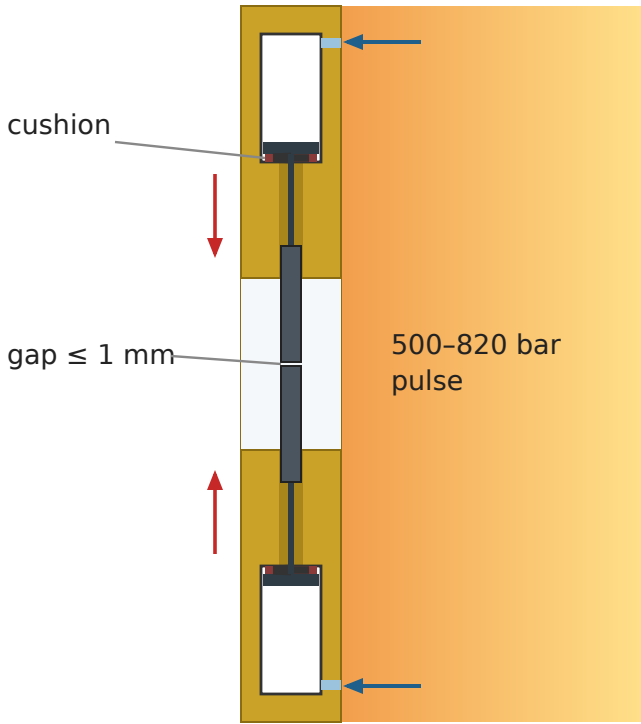


(a) Arrival: gate open



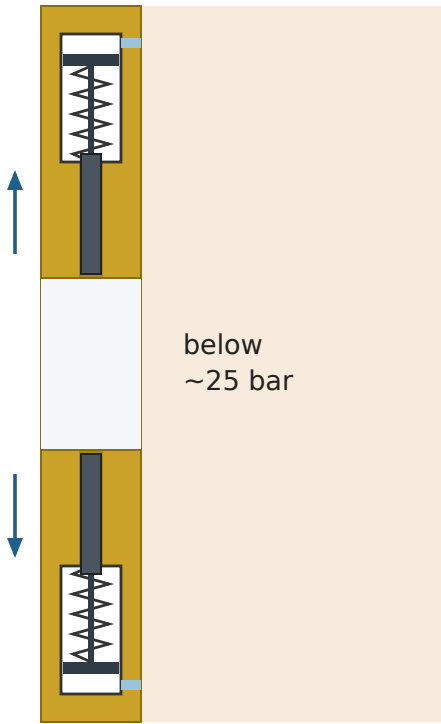
springs hold both leaves back;
the rod crosses the 10.9 cm port in 9 μ s

(b) Pulse: gate slams shut



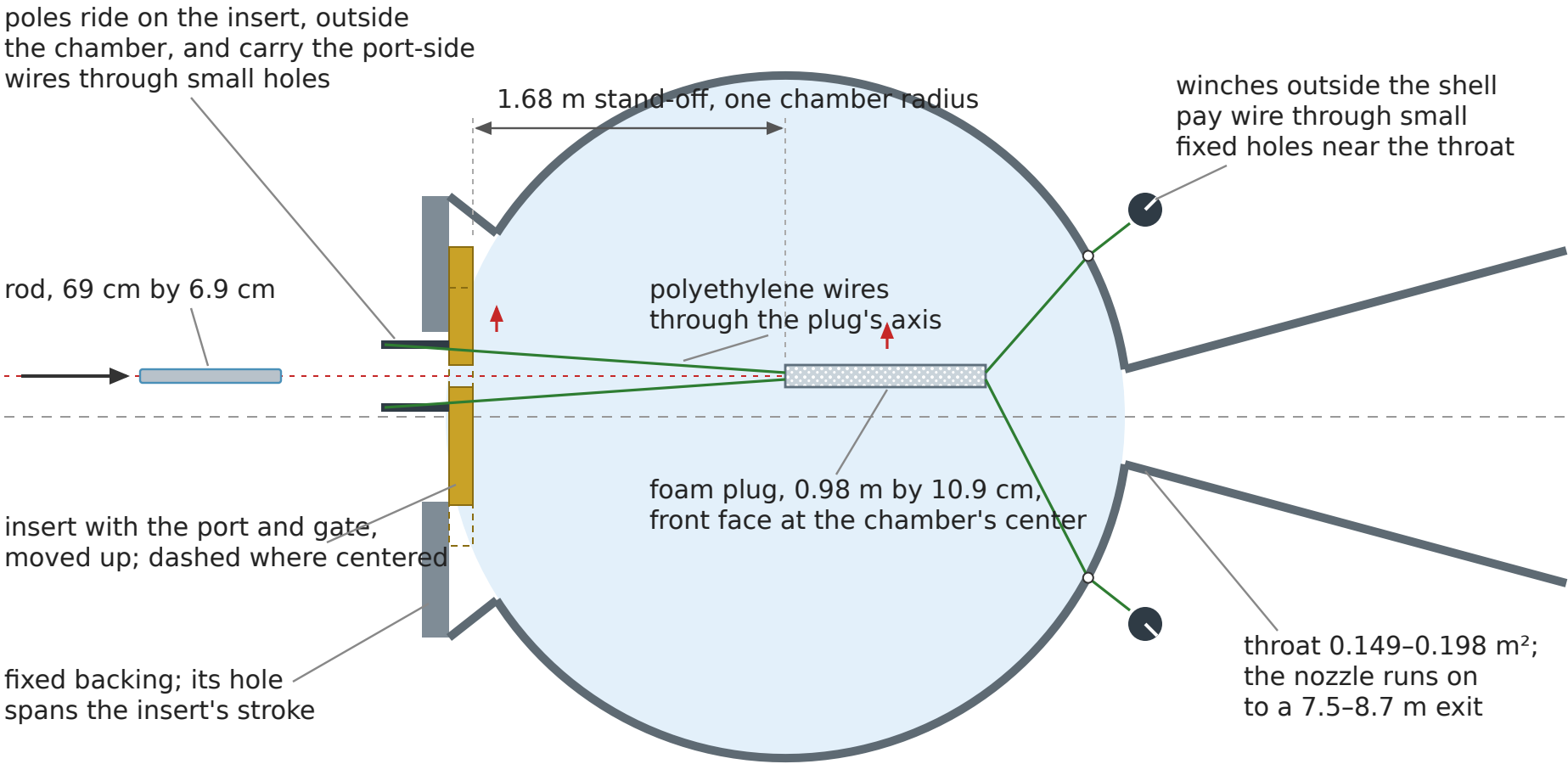
chamber gas drives the pistons (100 kN);
the leaves meet in 2-3 ms at 70-85 m/s

(c) Blowdown: springs reopen



a 5 kN spring preload wins once the
chamber falls below ~25 bar, 115-150 ms

(d) The plug hangs on wires and follows the port



20 m³ chamber, 1.68 m radius, filled with the charge gas

The insert's offset, the backing hole and the wire angles are exaggerated. The insert moves the port up to 10 cm;
the poles move with it, the winches pay out to match, and about 13 kN of tension holds the plug within 5 mm at 3 g.